A/B Testing

A/B Testing (also called **split testing**) is a **controlled experiment** used to compare **two versions of something** to determine **which one performs better** based on real user behavior.

It is widely used in **data science, product development, marketing, and AI systems**.

Idea:
A/B testing = show Version A to one group and Version B to another group, then measuring which performs better **using data.**

A B

Example (Website Without/With Chatbot)
Imagine you run a website.

Version	Chatbot	Users	Purchases
A	Without Chatbot	1000	80
B	With Chatbot	1000	120

Conversion Rate:
A → 80 / 1000 = **8%**
B → 120 / 1000 = **12%**

So, Version **B performs better**.

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Chatbot:
Hello. How can I help ?

Example (ML Model)

Suppose you built a **new ML model** for spam detection.
You now have:
Model A (Control) → Old production model (Logistic Regression)
Model B (Variant) → New model (Transformer / XGBoost)

We have 2 group of users: A and B.
Incoming emails are randomly divided and sent to these models:

User Group	Model Used	Traffic
Group A	Old Model	50%
Group B	New Model	50%

50% email → **Model A**

50% email → **Model B**

Metrics to Measure:

Instead of only accuracy, real business metrics matter:

Metric	Why Important
Spam detection rate	catches spam
False positives	avoids blocking real emails
User complaints	real satisfaction
Email open rate	downstream impact

50% email → **Model A**

50% email → **Model B**

Example Results After 1 Week

Metric	Model A	Model B
Accuracy	92%	94%
False Positive Rate	5%	2% (IMPORTANT)
User Complaints	120	45 (IMPORTANT)
Spam Detected	8,000	9,300 (IMPORTANT)

Model B clearly improves real outcomes.
So, deploy Model B to 100% users.

50% email → **Model A**

50% email → **Model B**

How A/B Testing Works (Step-by-Step)

1) Define Goal

Example:

- Increase signup rate
- Improve click rate
- Reduce bounce rate (Users leaving immediately)

2) Create Hypothesis

Example:

“Having a chatbot will increase conversions.”

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3) Split Users Randomly

Group A → sees old version, without chatbot

Group B → sees new version, with chatbot

(Randomization is VERY important.)

4) Run Experiment

Collect data for sufficient time.

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5) Analyze Results

Use statistical testing:

- t-test
- z-test
- chi-square test

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Statistical Concept Behind A/B Testing:
A/B testing relies on **hypothesis testing**.

- **Null Hypothesis (H_0):** No difference between A and B.
- **Alternative Hypothesis (H_1):** Version B is better than A.

If probability (p-value) is small,
-> Reject H_0.
-> There is statistical difference
between A and B

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6) Make Decision

Choose version with **statistically significant improvement**.

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Real-Life Uses

E-commerce:

- Product page layout
- Pricing display
- Checkout flow

Apps:

- UI design
- Notification timing
- Feature rollout

A

B

Real-Life Uses

Email Marketing:

- Subject lines
- Email design
- Send timing

Machine Learning:

- Compare two models in production

Why A/B Testing is Powerful

- Data-driven decisions and Removes guessing
- Improves products scientifically

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Key Business Metrics Used for web pages

Metric	Meaning
Conversion Rate	% users completing action
CTR	Click Through Rate
Bounce Rate	Users leaving immediately
Revenue per user	Business impact
Engagement Time	User interest

STOP

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